

The Global Biodiversity Standard:
Manual for assessment and best practices

Section 4

Remote Sensing Survey



A logged lowland dipterocarp forest under assisted natural regeneration in Sabah, Malaysia. (David Bartholomew)



The Global
Biodiversity
Standard

Section 4: Remote Sensing Survey

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The Global Biodiversity Standard (TGBS) uses both remote sensing and field survey data to understand performance of a site against the criteria of the standard. Remote sensing technology employs sensors mounted on drones, aircraft and satellites to remotely collect data about the characteristics of vegetation, enabling the analysis of ecological properties without direct contact. For the TGBS, the remote sensing survey supports the assessment process by monitoring over time the sub-attributes of the SER Five-star System to assess ecosystem integrity ([Appendix C](#)).

The TGBS uses remotely sensed data to:

- Complement field surveys to support assessments
- Increase spatial and temporal coverage of site assessments
- Monitor historical trends in ecosystem integrity sub-attributes and provide baseline data
- Provide comparisons between baseline conditions, current conditions, degraded sites and reference sites
- Improve accuracy for ecosystem integrity sub-attributes that are difficult to monitor during field surveys
- Verify field survey data

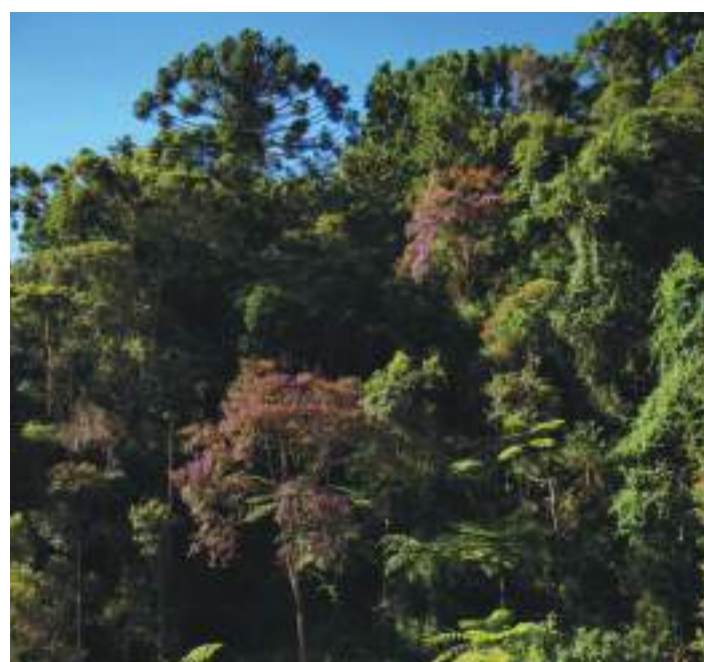
The TGBS assesses sites based on alterations in ecosystem integrity. This assessment relies on analysing time series data to establish a

comprehensive understanding of both historical baselines and current ecological conditions. Remote sensing data acquired by airborne and spaceborne sensors offer varying temporal resolutions; spaceborne sensors can systematically gather data at regular intervals, allowing for consistent monitoring over time. In contrast, drone and airborne sensor deployment typically requires specific surveys and is often event-driven or mission-specific, leading to less regular data acquisition (Figure 4.1). Consequently, the TGBS primarily uses spaceborne sensors, leveraging their ability to readily produce recurrent time series that facilitate the tracking of historical trends.

In addition to the analysis of spaceborne image time series, the use of unoccupied aerial vehicles (UAVs) to acquire high-resolution remote sensing data is recommended to support field surveys. Details on the use of UAVs are provided in [section 4.4](#).

The remote sensing survey shall be implemented by a specialist with experience of remote sensing data and analysis. The specialist may be located at one of the TGBS hubs or be an independent third party, depending on the capacity of the hubs. Such third-party organisations can work with multiple hubs.

The remote sensing survey shall typically precede the field survey. This allows for interesting signatures detected by the remote sensing survey to be investigated and verified during the field survey. In some cases, additional remote sensing surveys may follow field surveys if additional data is sought by the assessor.



Atlantic Forest, Itamonte, Brazil. (David Bartholomew)

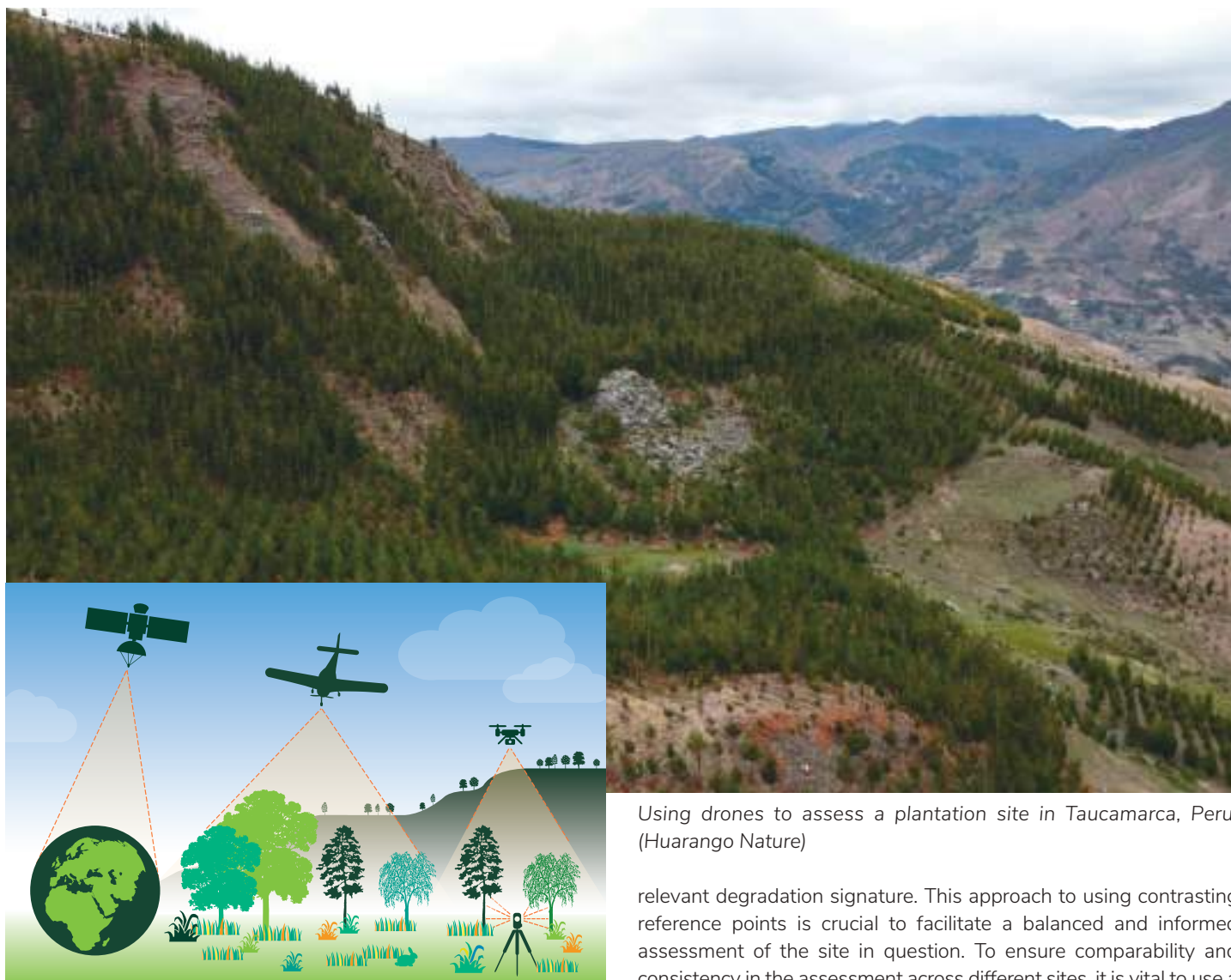


Figure 4.1 – Remote sensing data can be collected across a range of scales, including from sensors attached to spaceborne, airborne (planes and UAVs) and terrestrial devices. The Global Biodiversity Standard uses a range of remote sensing data to understand the ecosystem integrity of a site.

4.1 Preparing for the Remote Sensing Survey

Under criteria 1 and 4-7 of the TGBS, the evaluation of sites hinges on their restoration progress compared to a reference model. Remote sensing surveys are instrumental in this comparison, allowing for the assessment conditions to be juxtaposed with multiple reference sites that shape the reference model. In scenarios where pristine reference sites are unavailable, the most ecologically intact sites accessible shall be pinpointed as substitutes. Furthermore, it is essential to compare the project site not only with high-integrity reference sites but also with a highly degraded site that registers as zero stars within the Five-star System. Should a zero-star site be unattainable, the site with the lowest star rating shall be earmarked.

The identification of both highly intact and severely degraded sites is pivotal, and their selection must resonate with the specific degradation experienced by the project site to ensure accurate monitoring of the

Using drones to assess a plantation site in Taucamarca, Peru. (Huarango Nature)

relevant degradation signature. This approach to using contrasting reference points is crucial to facilitate a balanced and informed assessment of the site in question. To ensure comparability and consistency in the assessment across different sites, it is vital to use a single sensor for remote sensing surveys. This is because different sensors may have disparate spectral and spatial resolutions, which can introduce variability that complicates direct comparisons. The consistent use of one sensor type across all sites in an assessment (reference sites, degraded sites and the site under assessment) enables a standardised evaluation framework that accounts for the varying conditions of degraded and non-degraded sites alike. Assessors are advised to designate these reference and degraded sites with careful consideration before initiating the remote sensing survey to guarantee that the assessment is accurately benchmarked.

In addition, prior to conducting the remote sensing survey, the assessor shall evaluate the options available and how they can be used to support the assessment of a site. In particular, the assessor shall consider:

- Limitations with respect to spatial, temporal and sensor resolution of data
- Use a single remote sensing mission to assess reference and degraded trends
- The length of available time series
- The potential to increase spatial coverage of assessments (larger sites require more extensive remote sensing surveys to ensure sufficient coverage of the site)

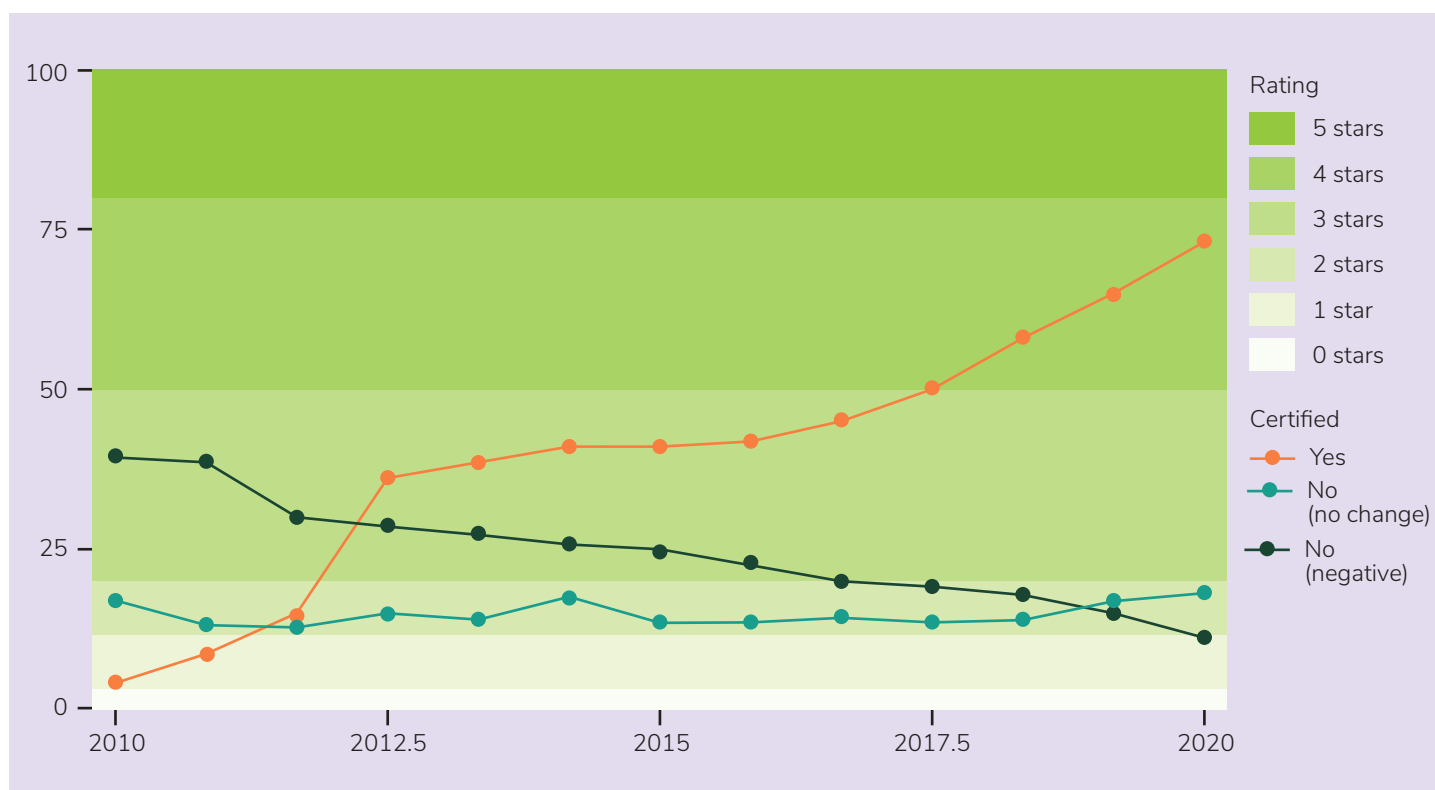


Figure 4.2 – A simulated example of a time series generated to quantify changes in productivity & cycling using the Normalised Difference Vegetation Index (NDVI). Data is standardised as a percentage relative to the reference (100%) and highly degraded (0%) sites using the analysis and compared with quantitative thresholds for the Five-star ratings for the sub-attribute. In this example, the site shows positive changes (orange) towards the reference model, and will score highly and attain TGBS certification. Projects that show no change (green) or decline relative to the reference model (black) will be awarded a low score that is insufficient to attain TGBS certification.

- The cost of data for remotely sensed products that are not open-source
- The appropriateness of methodologies to the assessment site
- The time and expertise required for data analysis
- The need to fill knowledge gaps from field surveys
- The potential to improve the accuracy of assessments
- The TGBS assessment requirements, such as the sub-attributes of ecosystem integrity that are more challenging to assess from field data alone
- The difficulty of field surveys because of challenging site conditions, e.g., inaccessibility to certain areas, remoteness, dangerous terrain, habitats of dangerous animals

Assessors shall discuss the options with the remote sensing specialist (if external to the hub) prior to the remote sensing survey. This consultation should be used to identify the highest priority, most reliable and cost-effective remote sensing options to be deployed.

4.2 Conducting the Remote Sensing Survey

The remote sensing survey for the TGBS includes comparing historical trends at the site under assessment against the reference model and highly degraded sites. The process of extracting historical trends involves using multitemporal satellite imagery to develop a comprehensive time series of biophysical characteristics of the assessment site, reference sites and highly degraded sites (Appendix D).

After generating time series for the three types of sites, the initial step involves identifying patterns of change in the remote sensing signal over time. In the case of reference sites, it is anticipated that variations in the remote sensing signal are predominantly due to natural seasonal fluctuations in vegetation properties, such as changes across growing seasons. These patterns observed in the reference sites serve as a baseline to evaluate the relative recovery of highly degraded sites. It is vital to ensure that both the reference and the highly degraded sites share similar edaphic and climatic conditions and encompass identical vegetation formations for accurate comparison and analysis. Sites under assessment that are near to each other and have the same edaphic and climatic conditions may share the same reference sites.

Upon generating the time series data, the subsequent step entails assessing the extent and direction of change in sub-attributes from the project inception to the present conditions. This analysis shall be conducted directly using the time series data of the project area. The outcomes of this analysis shall then be compared against the quantitative thresholds of the Five-star System, allowing for the determination of the site's star rating under both baseline and current conditions (Figure 4.2).

A spatial analysis of the data shall also be undertaken to identify areas with the biggest changes in sub-attribute values. Heat maps showing the magnitude of change can be useful ways to display the results and to identify areas with exceptional values.



Identification of over-utilisation using a drone in Taucamarca, Peru. (Huarango Nature)

A report with the results from the remote sensing survey shall be generated and provided to the assessors. The assessors shall use this report to compare results derived from the field and remote sensing surveys to assess the site ([section 6](#)).

4.3 Interpretation of Remote Sensing Data for Assessments

On completion of the remote sensing survey, the assessor shall discuss the results with the remote sensing specialist. This consultation shall be used to explain the analysis and results from the remote sensing survey. Whilst it may be useful to understand the technical details of the survey, the consultation should focus on facilitating an accurate interpretation of the results by the assessor. The remote sensing specialist shall explain any metrics used, including whether they are direct or indirect indicators of biodiversity. The remote sensing specialist shall also explain any limitations with the data that should be considered when making assessments.

The remote sensing survey can give strong indicators of the baseline conditions of the site because of its capacity to use data to directly measure the ecosystem attributes through time series. The remote sensing survey can be used to verify indirect indicators of baseline conditions that are identified through the field survey or to verify baseline monitoring data that may be provided by the applicant.

Remote sensing can provide novel insights to both baseline and current conditions that may not be discovered through the field survey because of methodological limitations. Some sub-attributes may be difficult to measure reliably during the field survey because methods are complex, expensive, require specific expertise or are time intensive. In instances when some field survey data may be unreliable, the remote sensing survey may represent a more reliable method for assessing the site for those data.

In addition to analysing overall trends, data generated from the remote sensing survey shall be analysed spatially. Heat maps are one useful tool to highlight spatial trends in the data. Spatial analysis of remotely sensed data is crucial for determining whether the locations selected for field surveys accurately represent the entire site. Ideally, the planning of field surveys shall be guided by remote sensing imagery to ensure comprehensive coverage of the site, as detailed in [section 5](#). Field surveys shall be used to validate and corroborate the findings obtained from the remote sensing analysis. The remote sensing survey will produce data and trends in ecosystem attributes. The data shall be inspected to identify clear trajectories in sub-attributes relative to the reference and highly degraded ecosystems. Strong positive trends can provide a clear indication that a sub-attribute is improving and recovering towards the reference model. In contrast, negative trends can indicate ongoing degradation, leading to negative scores being assessed for these sub-attributes at the site. Sites that do not show strong positive or negative trends indicate that little change has occurred at the site for the respective sub-attribute.

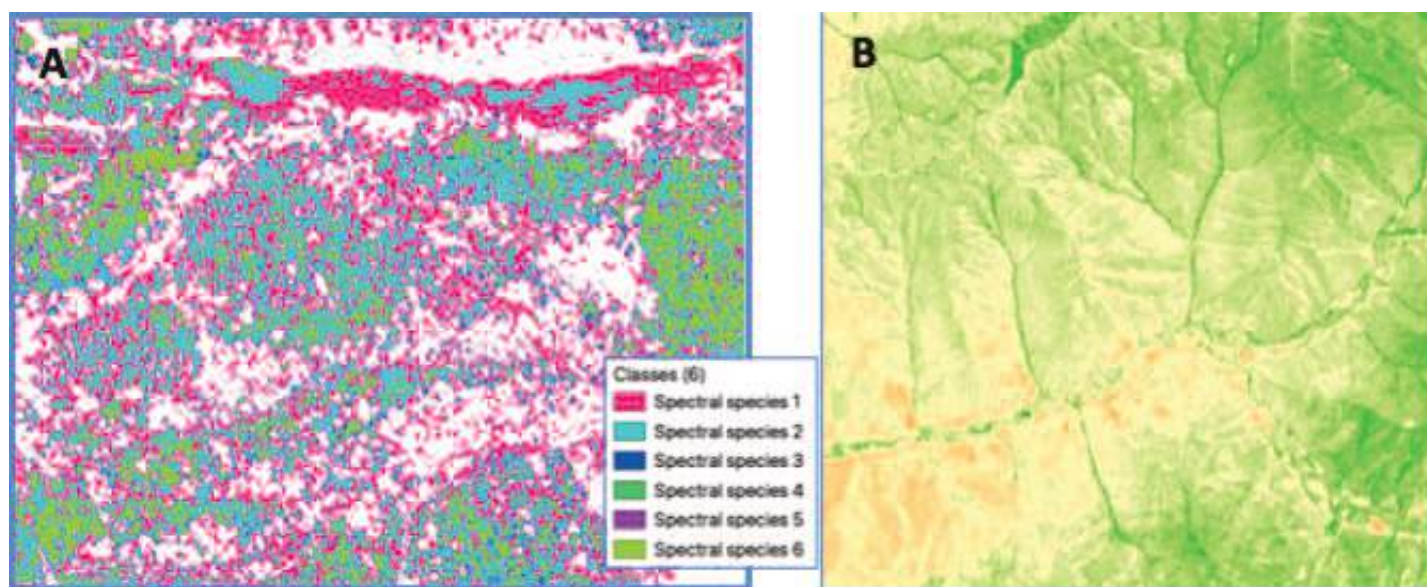


Figure 4.3 – Examples of metrics that can be used in the remote sensing survey: (A) Spectral species product that can be used to identify and spatially analyse canopy species diversity; (B) Vegetation indices that can be used to measure biomass. Source: Reforest'Action.

4.4 Use of Unoccupied Aerial Vehicles (UAVs) Equipped with Sensors in Field Surveys

Remote sensing technology, a key tool in the assessment toolkit of the TGBS, has undergone significant evolution with the development of unoccupied aerial vehicles (UAVs), commonly known as drones. This section explores the nuanced role of UAVs in the context of field surveys, supplementing traditional remote sensing methods to enhance the precision and efficiency of site assessments. However, it is essential to acknowledge the challenges and limitations associated with the widespread use of UAVs, particularly in regions where operational restrictions and difficulties may pose significant hurdles. Appropriate permissions must always be obtained prior to any use of UAVs in the TGBS assessment process.

In the preparatory phase, the inclusion of UAVs as a remote sensing tool introduces a dynamic dimension to site assessments. The identification of reference and highly degraded sites gains specificity, as UAVs allow for targeted and high-resolution data acquisition, aligning with the specific degradation signature experienced by the project site. The selection of a single UAV and sensor type for consistency in data collection minimises variability, ensuring standardised evaluations across diverse sites.

UAVs play a pivotal role in capturing high-resolution imagery, providing a detailed spatial and temporal perspective for the assessment of biophysical characteristics. A wide range of existing and emerging monitoring tools of restoration projects involve UAVs (de Almeida et al. 2019; McKenna et al. 2022; Robinson et al. 2022) and could be considered to support the field survey. The use of UAVs allows for a more flexible and targeted survey approach compared to traditional satellite imagery, and the detailed spatial and temporal information acquired facilitates the identification of patterns of change, aiding in the assessment of sub-attributes over time. The adaptability of UAV deployment, whether event-driven or mission-specific, adds a responsive layer to the assessment process.

The incorporation of UAVs in remote sensing surveys necessitates a collaborative effort in interpreting data between assessors and specialists. The results obtained from UAV-based surveys, integrated into the broader remote sensing dataset, offer nuanced insights. The ability of UAVs to capture high-resolution data aids in validating field survey locations and provides a detailed spatial understanding of ecosystem attributes. This spatial granularity is particularly crucial for assessing small-scale changes and validating trends observed in broader satellite imagery.

While UAVs offer unparalleled flexibility and targeted data acquisition, their use is not without challenges. Operational restrictions, regulatory frameworks and regional difficulties may impede the seamless integration of UAVs into field surveys. In areas with stringent regulations, obtaining the necessary permits for UAV deployment might be a time-consuming and bureaucratic process, potentially delaying the assessment timeline. Moreover, local topography and weather conditions could pose operational difficulties for UAV flights, limiting their applicability in certain regions. Also, the constant evolution of technology in both UAV platforms and deployed sensors may cause difficulties in maintaining temporal observations with the same parameters.

The adaptability of UAV deployment, while advantageous, is contingent on the regulatory landscape. Some regions of the world may have restrictions on flying UAVs near sensitive areas, such as protected areas and wildlife sanctuaries or even populated zones, restricting the comprehensive coverage needed for a thorough field survey. Additionally, concerns about privacy and data security may further complicate UAV operations, necessitating careful navigation of legal and ethical considerations. Consequently, the TGBS does not stipulate the required use of UAVs during the field survey of sites.

4.5 Use of Remote Sensing for Reassessment

Five years after the award of the Global Biodiversity Standard, a site must be reassessed before the certification can be renewed. A remote sensing survey must be done for all reassessments of a site to confirm that improvements in ecosystem integrity have been maintained since the previous assessment.

In addition to the remote sensing survey outlined in sections 4.1–4.4, remote sensing shall be used in the reassessment process to identify major changes that have occurred since the previous field survey. An analysis of land cover change and land productivity between field surveys at the site can highlight important areas that should be visited by the field survey team during the field survey required for reassessment. Large changes in land cover or productivity could identify areas that have become degraded or have had intensive management since the previous assessment. These areas could highlight both positive and negative activities implemented at the site since the previous assessment.



Training on how to use a drone for TGBS assessments.
(Teresiah Mungai)



Ankafobe reference forest as viewed from a drone.
(Toky Ralainaorina)

When employing remote sensing imagery for the reassessment of a certified site, it's crucial to adhere to the consistent methodology established in [section 4.2](#). This involves using the same remote sensing mission that was used in constructing the time series analysis over the area, ensuring a reliable comparison of data across different time periods. Additionally, it is essential to account for critical factors that may affect the analysis, such as the type of atmospheric correction method used. By maintaining this consistency, the remote sensing approach can more accurately detect and quantify significant changes in land cover and productivity, as identified in the initial survey. This rigorous approach is vital for pinpointing areas that have undergone considerable ecological transformation since the last assessment, whether it be degradation or improvement.

4.6 Limitations of Remote Sensing

Remote sensing technology provides an important monitoring tool for the Global Biodiversity Standard and can give insights to the historical trends in biodiversity that may not be possible to achieve through field surveys alone. Despite the many advantages that remote sensing provides, there are several limitations that assessors should consider when interpreting and using remotely sensed data. These limitations shall be identified and considered before assessments are made. Here we present some of the most common limitations associated with remotely sensed data.

4.6.1 Indirect Measures of Ecosystem Attributes

Remote sensors on board satellites or aeroplanes measure the interaction of electromagnetic radiation with the Earth's surface. By quantifying this interaction, it becomes possible to infer the biophysical and biochemical properties of vegetation. These properties are closely linked to key ecosystem processes, including carbon sequestration and biodiversity. However, it's important to note that these estimates often tend to be conservative and require field validation.

Indirect measures of biodiversity and ecosystem attributes may also provide unreliable data if different targets give similar results. For example, some species may have similar spectral signatures that arise

from similar traits and therefore they may be interpreted as the same species, resulting in lower than expected species diversity for the site (Rocchini et al. 2022). Moreover, it may not be possible to identify whether changes in an index reflect positive or negative changes with respect to biodiversity. This is a particular limitation for methods that are unable to distinguish between native and non-native species. The sensitivity of remote sensing to measure biodiversity improves with the spectral resolution, which is covered in detail in [section 4.6.3 Spectral Resolution](#).

4.6.2 Spatial Resolution

The spatial resolution of remote sensing refers to the size of the smallest object that can be reliably identified and distinguished in an image captured by a remote sensing instrument, such as a satellite or an aerial camera. Associated to the spatial resolution on an image is the ground sampling distance (GSD), which indicates the actual ground distance, in metres or centimetres, represented by each image pixel. For instance, a GSD of 1 metre means that each pixel in the image represents a square of 1 metre by 1 metre on the ground. Due to technological limitations inherent in satellite remote sensing, the spatial resolution of data acquired from spaceborne sensors is currently capped. Low spatial resolution may affect the ability to detect features in the landscape and can be a particular challenge for the monitoring of biodiversity. Spatial resolution is a particular limitation for many spaceborne sensors when compared to airborne and ground-based technologies.

The spatial resolution of the sensor used in the remote sensing survey should be compared with the size of the organism or feature being monitored to understand if the spatial resolution of the data is sufficient. In cases when the spatial resolution is lower than the size of the organism or feature, data will represent an average for the pixel, which could lead to both over- and underestimation of variables.

4.6.3 Spectral Resolution

Spectral resolution is characterised by the quantity and the bandwidth of the spectral bands that a remote sensing instrument can capture. Multispectral sensors typically capture images using a small number of broad spectral bands, usually fewer than five, each with a bandwidth greater than 50 nanometres, and these bands are non-continuous. In contrast, hyperspectral sensors are designed to capture images across hundreds of continuous, narrow spectral bands, each less than 10 nanometres in width. This feature of hyperspectral sensors enables them to provide highly detailed information about the spectral response of the Earth's surface, capturing subtle variations that multispectral sensors might miss.

For biodiversity assessments the spectral resolution plays a key role. Hyperspectral sensors are particularly effective in assessing biodiversity due to their ability to capture detailed spectral information across hundreds of narrow, continuous bands. This fine-scale spectral resolution allows for the precise discrimination of different species of vegetation and other surface materials. Each species has a unique spectral signature – a distinct pattern of reflection and absorption across various wavelengths – which can be accurately identified with hyperspectral data.



Measuring a tree to ground truth remote sensing data and ensure accuracy. (David Bartholomew)

4.6.4 Temporal Resolution, Revisit Time and Longevity

Temporal resolution in remote sensing refers to the frequency at which a sensor or satellite system captures images of the same area on the Earth's surface over time. Temporal resolution and revisit time, while related, are distinct concepts in remote sensing, particularly for very high resolution (VHR) sensors (with a GSD of less than 1 metre). Temporal resolution refers to the frequency at which a sensor can capture images of the same area over time. It's a broader term encompassing the sensor's ability to monitor changes at regular intervals. Revisit time, on the other hand, specifically denotes how frequently a satellite can physically pass over and record data from the same location.

VHR sensors often have advanced capabilities, such as off-nadir imaging, which allow them to capture images of areas not directly beneath their orbital path. This capability significantly enhances their ability to revisit specific areas more frequently than their standard orbital cycle would suggest. As a result, while the inherent revisit time of a sensor might be limited by its orbital parameters, the actual temporal resolution can be improved through off-nadir imaging capabilities. However, it's important to note that images obtained at large off-nadir angles (exceeding 20 degrees) may exhibit distortion issues. These distortions can affect the clarity and accuracy of object recognition within the images, thus impacting their overall utility in certain applications.

While VHR sensors offer enhanced capabilities like off-nadir imaging to frequently capture images of specific areas beyond their standard orbital cycles, there are inherent limitations in the temporal resolution of remote sensing products. The revisit period for

spaceborne sensors can vary significantly, ranging from days to years, which might not always align with the timing requirements of a project or its corresponding field survey. Moreover, certain conditions can hinder data collection during these revisits. For instance, passive optical sensors are unable to penetrate clouds or operate without daylight, and depend on daytime conditions for effective data acquisition. In areas with regular cloud cover, this dependence can lead to lower temporal resolution and intermittent data, causing potential gaps in monitoring.

The temporal resolution of time series generated from the remote sensing survey may be longer or shorter than the timeline for the project applying for certification. A time series longer than the project may capture impacts that were caused by the previous management activities at the site, whilst a shorter time series may miss impacts that were implemented in the earlier stages of the project. The assessor should compare the dates for the remote sensing time series with the project timeline to ensure appropriate interpretation of results of the remote sensing survey.

Limitations around sensor longevity can also limit the capacity for sites to be remotely sensed. The mission time for spaceborne sensors may not encompass the time of the management activities at the site. Like limitations around temporal resolution, a sensor launch date that post-dates the initiation of the project may cause the remote sensing survey to miss impacts that were implemented beforehand.

Combining data from multiple sensors is one option for overcoming spatial coverage and temporal limitations of spaceborne remote sensing of a site. Whilst this can be a useful method, the assessor should be aware that sensors often have different specifications, and therefore resolutions that can result in data that may not be fully comparable.

4.6.5 Seasonal Changes

Seasonal variations in vegetation, driven by phenological changes, significantly impact time series of remotely sensed data. Plant phenology refers to the seasonal variation in plant characteristics, and it may alter the spectral signatures captured in remote sensing imagery over time. For example, the increase in green leaf cover during spring enhances the reflectance in the visible and near-infrared wavelength range, which is captured by sensors. Conversely, autumnal leaf falls or vegetation dormancy leads to decreased reflectance. Such seasonal dynamics are crucial for understanding ecosystem processes, but they also introduce variability that must be accounted for when analysing remote sensing time series data for monitoring biodiversity.

4.6.6 Cost

Remote sensing surveys can be expensive to implement because of the costs of acquiring and processing data, maintenance of equipment and technical expertise. Assessors should therefore carefully consider the priorities for the remote sensing survey, the additional data and evidence the remote sensing survey can provide, and the costs involved for different remote sensing options.